Alexander Salganik

PhD Student in Astrophysics

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Last CV Update: Monday 29th September, 2025

Summary

I am a highly motivated PhD student at the University of Turku with five first-author publications in *MNRAS* and *A&A*. My research focuses on X-ray Astrophysics, particularly on X-ray pulsars. I have extensive experience in spectral and timing analyses using data from leading space missions and observatories. My primary goal is to expand my research to include X-ray polarimetric observations of pulsars, aspiring to become a world-leading researcher or professor in the field of high-energy astrophysics.

Education

St. Petersburg State University 2018–2024
Integrated Bachelor's and Master's Degree in Astrophysics (with Distinction)

University of Turku 2024–present
PhD in Astrophysics (Self-funded research supported by EDUFI and Wihuri grants)

Work Experience

Space Research Institute (IKI RAS) 2019–2023
Senior Laboratory Assistant (part-time)

University of Turku 2025
Teaching Assistant/Demonstrator – Observational X-ray and Gamma-ray Astrophysics Course

Professional Service

- Peer-review for *The Astrophysical Journal* 2024–present
- Peer-review for *Advances in Space Research* 2024–present
- Member of the Finnish Astronomical Society (FAS) 2025–present
- NewAthena Science Community Member 2025–present

Publications — First Author

1. ***On the nature of the X-ray pulsar XTE J1859+083 and its broadband properties*** 2022
   Salganik A., Tsygankov S. S., Djupvik A. A., Karasev D. I., Lutovinov A. A., Buckley D. A. H., Gromadzki M., Poutanen J.
   MNRAS 509, 5955. DOI
2. ***Discovery of a pulse-phase-transient cyclotron line in the X-ray pulsar Swift J1808.4−1754 and identification of an optical companion*** 2022
   Salganik A., Tsygankov S. S., Lutovinov A. A., Djupvik A. A., Karasev D. I., Molkov S.V.
   MNRAS 514, 2707. DOI

3. ***First characterization of a new high-mass X-ray binary in LMC eRASSU J050810.4−660653 with SRG/ART-XC, NuSTAR, and Swift*** 2022

   Salganik A., Tsygankov S. S., Lutovinov A. A., Mushtukov A. A., Mereminskiy I. A., Molkov S.V., Semena A.N.
   MNRAS 514, 4018. DOI

4. ***RX J0440.9+4431: another supercritical X-ray pulsar*** 2023

   Salganik A., Tsygankov S. S., Doroshenko V., Molkov S.V., Lutovinov A. A., Mushtukov A. A., Poutanen J.
   MNRAS 524, 5213. DOI

5. ***Discovery of a bimodal luminosity distribution in persistent Be/X-ray pulsar 2RXP J130159.6-635806*** 2025

   Salganik A., Tsygankov S. S., Chernyakova M., Malyshev D., Poutanen J.
   A&A 698, A71. DOI

Grants

- **EDUFI Fellowship**: Highly competitive fellowship supporting doctoral research (€17,100) — Oct 2024
- **Wihuri Foundation Grant**: Highly competitive grant supporting doctoral research (€30,000 total) — Oct 2024
- **Frontiers in X-ray Polarimetry (FiXP) Academy travel grant** (€1200) — Mar 2025

Awards and Honors

- Personal Scholarship of the Government of St. Petersburg in the field of Physics — 2018–2020
- IKI RAS Best Scientific Work Prize (for researchers under 35) — 2023
- Enhanced Academic Scholarship of St. Petersburg State University for academic and research excellence — 2020–2024

Accepted Proposals

General Observer:

- **PI: IXPE proposal for GX 1+4 observations.** — 2024
- PI: Insight-HXMT proposal for observations of cyclotron line harmonics. — 2024
- Co-I: IXPE proposal for 4U 1907+09 observations. — 2023

Director's Discretionary:

- PI: Swift 6 ks TOO observations of Swift J1808.4−1754 — Aug 2021
- PI: NuSTAR 56 ks TOO observations of eRASSU J050810.4−660653 — Dec 2021
- PI: Swift 39 ks TOO observations of eRASSU J050810.4−660653 — Dec 2021
- PI: NuSTAR 59 ks TOO observations of LS V +44 17 — Jan–Apr 2023
- PI: Swift 7 ks TOO observations of LS V +44 17 — Jan 2023
- PI: NICER 39 ks TOO observations of IGR J06074+2205 — Oct 2023
- PI: NICER 49 ks TOO observations of RX J0520.5−6932 — Apr–Aug 2024
- PI: Swift 10 ks TOO observations of AX J1910.7+0917 — May–Jun 2024
- PI: Swift 33 ks TOO observations of RX J0520.5−6932 — May–Aug 2024
- PI: Swift 7 ks TOO observations of V0332+53 — Oct 2024

- PI: Swift 22 ks TOO observations of 2RXP J130159.6-635806 — Sep 2024–Feb 2025
- PI: Swift 0.5 ks TOO observations of RX J0032.9−7348 — Nov 2024
- PI: NICER 2 ks TOO observations of RX J0032.9−7348 — Nov 2024
- PI: NuSTAR 140 ks TOO observations of SXP 31.0 — May 2025

Press & Public Outreach

- INTEGRAL Picture of the Month: Observations of RX J0440.9+4431 in the supercritical accretion regime — Jun 2023
- IKI RAS Press Release: RX J0440.9+4431 switched to the super-critical accretion regime — Jun 2023
- Main RAS Press Release: RX J0440.9+4431 transitions into super-critical accretion — Jun 2023

Talks

On the nature of the X-ray pulsar XTE J1859+083 and its broadband properties:

- Conference Fundamental and Applied Space Research 2021, IKI RAS — Apr 2021
- Invited Talk at St. Petersburg State University — Dec 2021
- Conference Astronomy and Space Exploration, Ural Federal University — Jan 2022

Study of the X-ray pulsars Swift J1808.4-1754 and eRASSU J050810.4-660653 based on space observatories' data:

- Conference Fundamental and Applied Space Research 2023, IKI RAS — Apr 2023

Study of the transition of the pulsar RX J0440.9+4431 to the supercritical accretion regime:

- Conference Physics of Neutron Stars 2023, Ioffe Institute — Jul 2023
- Invited Talk, Princeton University — Sep 2023
- Invited Talk, Ioffe Institute — Sep 2023
- Conference High-energy Astrophysics Today and Tomorrow HEA-2023, IKI RAS — Dec 2023

Discovery of a bimodal luminosity distribution in persistent Be/X-ray pulsar 2RXP J130159.6-635806:

- Invited Talk, THEAPA Group Meeting, University of Maryland — Mar 2025

Workshops

- **Frontiers in X-ray Polarimetry (FiXP) Academy** — Apr 2025